PRE-ALGEBRA (7TH) · UNIT 12: DATA ANALYSIS AND PROBABILITY

Board Game Inventor

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students invent a family board game. Spinners, dice, and a deck produce simple and compound probabilities, counting methods count outcomes and hands, and playtest data checks experimental probability against theory.

Students use probability (and expected value in Honors) to make their game fair.

Standards Addressed

- 7.SP.C.5–7 Probability models; theoretical vs. experimental
- 7.SP.C.8 Compound events: lists, tables, tree diagrams
- 7.SP.A/B Data displays and comparison
- S-CP / S-MD (Honors): permutations, combinations, expected value

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a classic board game. Part 1 basic probability.
Day 2	Part 2 counting and compound events.
Day 3	Part 3 playtests and rule book.

Materials

- Student handout and Playtest card
- Dice, paper spinners, and index cards
- Poster board for the game
- Calculators

How the Three Versions Differ

Level	What changes
CP2	Single-event probabilities, counting principle, two-dice sums, experimental probability, and a bar graph.
CP1	Independent and dependent events, permutations and combinations, and a fairness check.

Level	What changes
Honors	Complements, three-draw dependent events, combination probabilities, expected value, and rule changes for fairness.

Watch For (Common Misconceptions)

- Adding probabilities that should be multiplied.
- Forgetting to reduce the total after a card is drawn without replacement.
- Using permutations when order does not matter.

Extension Ideas

- Play other groups' games and rate their fairness.
- Run 200 simulated rolls with a spreadsheet.
- Design a game box with a probability fact.

Answer Key – CP2

Part 1: Chance Basics

#	Task	Answer
1	What is the probability the spinner lands on gold?	$\frac{1}{4}$ = 25%
2	What is the probability of drawing a bonus card?	$\frac{6}{20} = \frac{3}{10}$
3	What is the probability of NOT drawing a lose card?	$\frac{18}{20} = \frac{9}{10}$
4	Roll one die. What is the probability of rolling an even number?	$\frac{3}{6} = \frac{1}{2}$

Part 2: Counting Outcomes

#	Task	Answer
5	How many outcomes are there for one spin and one die roll?	4 × 6 = 24
6	What is the probability of spinning red and rolling a 6?	$\frac{1}{24}$
7	How many outcomes are there for two dice? What is the probability of a sum of 7?	36; $\frac{6}{36} = \frac{1}{6}$
8	Compare: in the playtest, a sum of 7 came up 9 times in 60 rolls. What is the experimental probability?	$\frac{9}{60} = \frac{3}{20}$ = 15% (theoretical ≈ 16.7%)

Part 3: Invent the Game

#	Task	Answer
9	Make a bar graph of the playtest results.	Bars: 5, 12, 26, 12, 5
10	Write a rule that uses the spinner and the deck. Find the probability that your rule happens on a turn.	Answers vary; e.g., "spin gold and draw bonus": $\frac{1}{4} \times \frac{3}{10} = \frac{3}{40}$.
11	Draw your game board and write three rules.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why might experimental probability be different from theoretical probability?	Random results vary in a small number of trials; more trials usually get closer.
R2	What makes a game fair?	Each player has the same chance of winning.

Answer Key – CP1

Part 1: Compound Events

#	Task	Answer
1	Find P(sum of two dice is 8 or more).	$\frac{15}{36} = \frac{5}{12}$
2	Find P(spin red and draw a bonus card).	$\frac{1}{4} \times \frac{6}{20} = \frac{3}{40}$
3	Draw two cards without replacement. Find P(both bonus).	$\frac{6}{20} \times \frac{5}{19} = \frac{30}{380} = \frac{3}{38}$
4	Draw two cards with replacement. Find P(both bonus). Why is it different?	$\frac{3}{10} \times \frac{3}{10} = \frac{9}{100}$; the first card is put back, so the second probability does not change.

Part 2: Counting

#	Task	Answer
5	In how many orders can 4 players take turns?	4! = 24
6	There are 8 tokens. In how many ways can 4 players each choose a different token (order matters)?	${}_8P_4 = 8 \times 7 \times 6 \times 5 = 1{,}680$
7	A player chooses 3 of the 8 bonus powers (order does not matter). How many choices?	${}_8C_3 = 56$
8	Explain why the token question is a permutation and the powers question is a combination.	Tokens go to specific players (order matters); powers are a set (order does not matter).

Part 3: Test & Fix

#	Task	Answer
9	Compare the experimental probability of rolling 6–8 with the theoretical probability.	Experimental $\frac{26}{60} \approx 43\%$; theoretical $\frac{16}{36} \approx 44\%$
10	In the current rules, Player 1 wins on doubles and Player 2 wins on a sum of 7. Is that fair? Use theoretical probabilities.	Doubles $\frac{6}{36}$, sum 7 $\frac{6}{36}$; equal, so it is fair. (Playtest: 11 vs. 9.)
11	Write your rule book with at least three rules, each with its probability, and explain why your game is fair.	Answers vary.

Reflection

#	Prompt	Look for
R1	How can you tell whether two events are independent?	If the outcome of the first does not change the probability of the second.
R2	Why are more playtests better?	Experimental probabilities get closer to theoretical ones as the number of trials grows.

Answer Key – Honors

Part 1: Probability Engine

#	Task	Answer
1	Find P(at least one 6 when rolling two dice).	$1 - (\frac{5}{6})^2 = \frac{11}{36}$
2	Draw three cards without replacement. Find P(no lose cards).	$\frac{18}{20} \times \frac{17}{19} \times \frac{16}{18} = \frac{68}{95} \approx 0.716$
3	Find P(exactly one bonus card in two draws without replacement).	$2 \times \frac{6}{20} \times \frac{14}{19} = \frac{168}{380} = \frac{42}{95}$
4	Is "spin gold" independent of "roll a sum of 7"? Show that P(both) = P(gold) × P(7).	Yes: $\frac{1}{4} \times \frac{1}{6} = \frac{1}{24}$; the spinner and dice do not affect each other.

Part 2: Counting & Probability

#	Task	Answer
5	A hand is 4 cards from the 20-card deck. How many hands are possible?	${}_{20}C_4 = 4{,}845$
6	How many hands contain both lose cards?	${}_2C_2 \times {}_{18}C_2 = 153$
7	Find the probability that a hand contains both lose cards.	$\frac{153}{4{,}845} \approx 0.032$
8	Five finish positions are awarded to 5 players. What is the probability that the youngest player finishes first and the oldest finishes last?	$\frac{3!}{5!} = \frac{6}{120} = \frac{1}{20}$

Part 3: Expected Value & Fairness

#	Task	Answer
9	Find the expected points per card drawn.	$\frac{8(2)+6(5)+4(0)+2(-8)}{20} = \frac{30}{20} = 1.5$
10	Player 1 earns 3 points for doubles and Player 2 earns 1 point for any sum of 6–8. Find the expected points per roll for each. Is it fair?	P1: $\frac{6}{36} \times 3 = 0.5$. P2: $\frac{16}{36} \times 1 \approx 0.44$. Not fair; P1 is favored.
11	Change one rule so the expected points are equal, and use the playtest data to support your choice.	e.g., P2 earns $\frac{9}{8}$ points, or P1 earns $\frac{8}{3}$; check with expected value. Playtest: P1 11 × 3 = 33, P2 26 × 1 = 26.

Reflection

#	Prompt	Look for
R1	How does expected value help design a fair game?	It shows the average points each player gains per turn; equal expected values make the game fair in the long run.
R2	When should you use the complement to find a probability?	When "at least one" is asked, since the complement "none" is easier to compute.